Business Problem and Analysis

Business Problem: **To determine whether the redesigned checkout page increases purchase conversion compared with the existing page.**

The e-commerce company wanted to determine whether the redesigned checkout page (Group B) increased purchase conversion compared with the existing page (Group A), and whether the improvement was statistically significant.

1. Conversion Rate Comparison

We calculated the purchase conversion rates for Group A and Group B to identify which checkout page converts more users into customers. Group A achieved a conversion rate of 5.03%, while Group B achieved 5.64%, indicating better observed performance by the redesigned checkout page.

2. Hypothesis Testing

We performed a two-proportion z-test to determine whether the difference in conversion rates was statistically significant. The p-value was 0.0194, below the 0.05 significance level, so we rejected the null hypothesis and concluded that the conversion rates were significantly different.

3. Conversion Uplift

We calculated the absolute and relative conversion uplift to measure the improvement achieved by Group B compared with Group A. Group B showed an absolute increase of 0.61 percentage points and a 12.05% relative improvement in conversion rate.

4. 95% Confidence Interval

We calculated a 95% confidence interval to estimate the plausible range of the conversion-rate improvement. The interval was 0.10 to 1.12 percentage points. Since it excludes zero, it supports the conclusion that Group B's conversion rate was significantly higher than Group A's.

Final Business Conclusion

The results suggest that the redesigned checkout page (Group B) performed better than the existing page (Group A), with a 12.05% relative improvement in conversion rate. The difference was statistically significant, and the confidence interval supports a positive improvement. Therefore, Group B is a promising option for rollout after validating experiment quality and business impact.